Practice workbook · Identify arteries, veins and capillaries and relate their structure to function

Comparing and explaining vessel structures, capillary adaptations for exchange, and the jobs of particular vessels.

17 question parts: 1 worked-example retry and 16 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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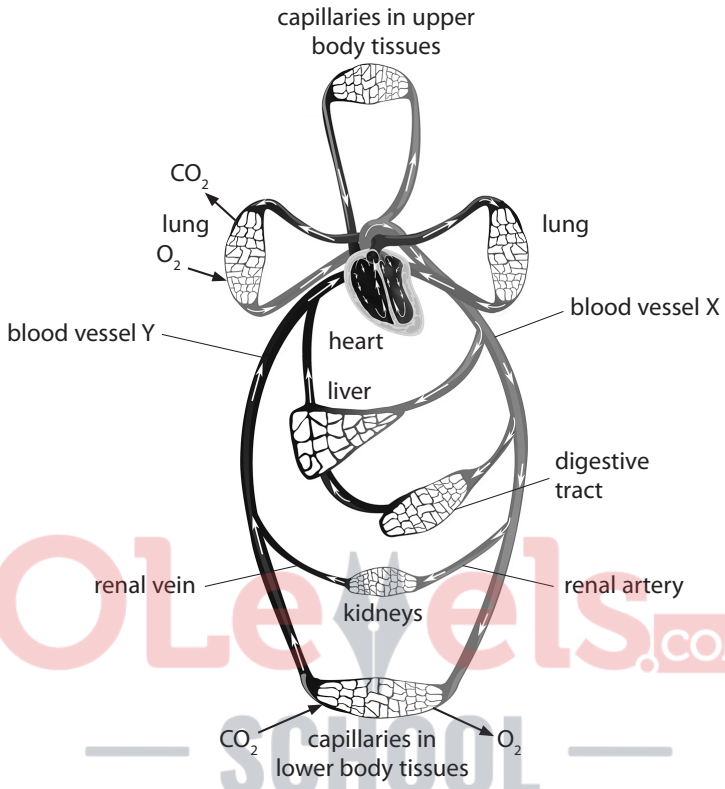
Type ID: HBI-RELATE-VESSEL-STRUCTURE-TO-FUNCTION

1. Question 6 · part a.iv · 3 marks

June 2019 Paper 2

Worked-example retry · Attempt part a.iv only.

6 The diagram shows the human circulatory system.



(Source: © Designua/Shutterstock)

(a) (i) What is the name of blood vessel X?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

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1. Question 6 · part a.iv · 3 marks

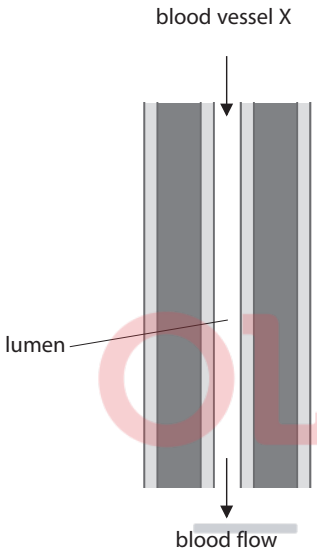
June 2019 Paper 2

Worked-example retry · Attempt part a.iv only.

(iii) The diagram shows a longitudinal section through blood vessel X.

Draw a diagram to show a longitudinal section through blood vessel Y.

(3)



(iv) Explain the differences in the structures of blood vessel X and blood vessel Y.

(3)

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1. Question 6 · part a.iv · 3 marks

June 2019 Paper 2

Worked-example retry · Attempt part a.iv only.

(b) Explain two adaptations of capillaries that increase the rate of diffusion of substances into body tissues.

(4)

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(Total for Question 6 = 12 marks)

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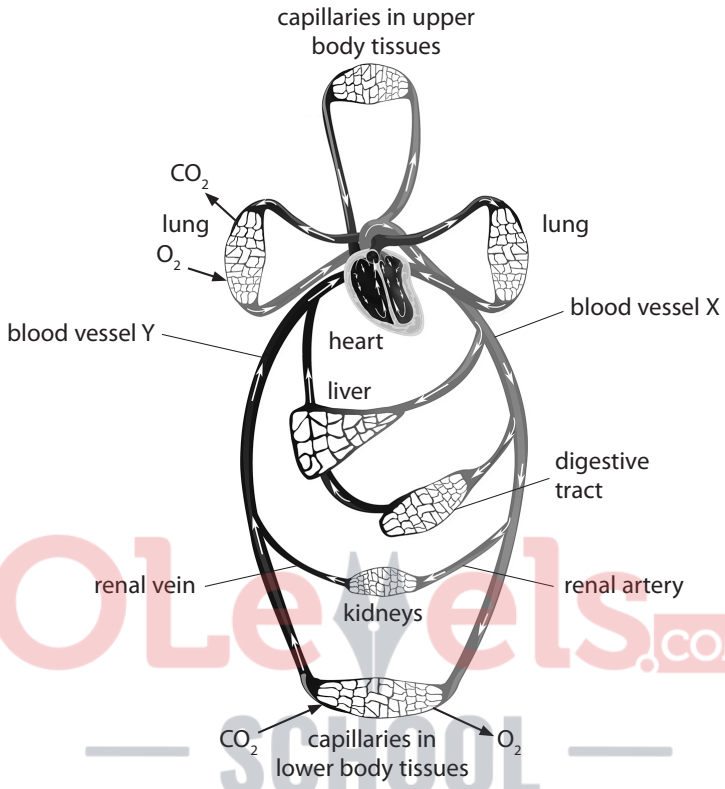
Working and answer

2. Question 6 · part a.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

6 The diagram shows the human circulatory system.



(Source: © Designua/Shutterstock)

(a) (i) What is the name of blood vessel X?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

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2. Question 6 · part a.iii · 3 marks

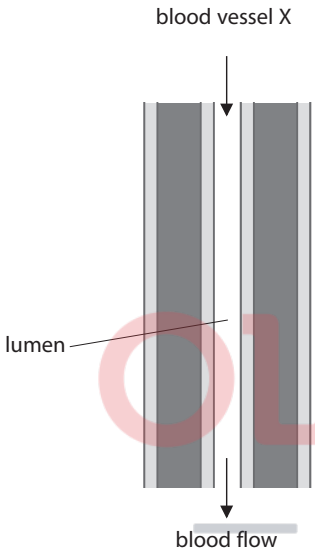
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Further practice · Attempt part a.iii only.

(iii) The diagram shows a longitudinal section through blood vessel X.

Draw a diagram to show a longitudinal section through blood vessel Y.

(3)



(iv) Explain the differences in the structures of blood vessel X and blood vessel Y.

(3)

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2. Question 6 · part a.iii · 3 marks · Working and answer

Working and answer

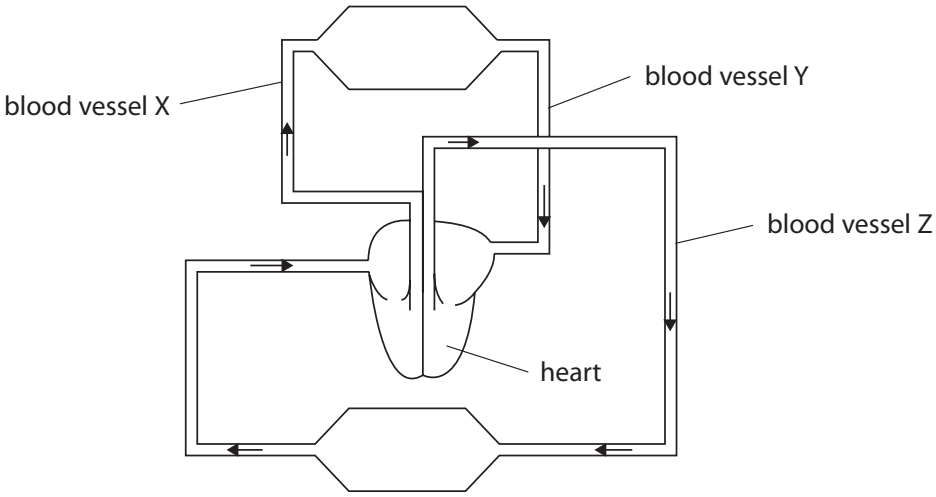
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3. Question 1 · part c · 3 marks

June 2022 Paper 2

Further practice · Attempt part c only.

1 The diagram shows a simple version of the human circulatory system.



(a) Name blood vessel X. **(1)**

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3. Question 1 · part c · 3 marks

June 2022 Paper 2

Further practice · Attempt part c only.

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(b) The box gives words about the circulatory system.

body	oxygenated	lungs	ventricle
	deoxygenated	atrium	

Complete the sentences using words from the box.

(5)

Blood vessel Y transports ..... blood from the

..... to the left ..... of the heart.

Blood vessel Z transports blood from the left ..... of the heart

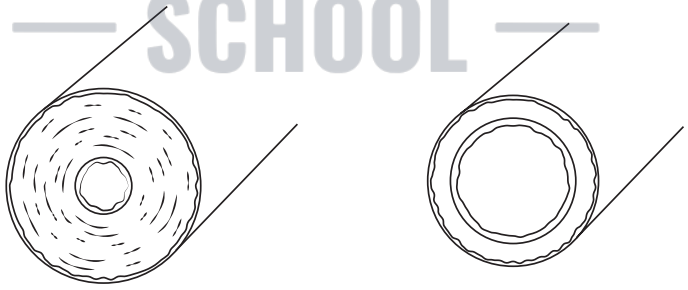
to the .....

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(c) The diagram shows the structure of blood vessel X and blood vessel Y.



blood vessel X blood vessel Y

Compare the structure of blood vessel X with the structure of blood vessel Y.

(3)

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(Total for Question 1 = 9 marks)

3. Question 1 · part c · 3 marks · Working and answer

Working and answer

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4. Question 2 · part b · 2 marks

June 2024 Paper 1R

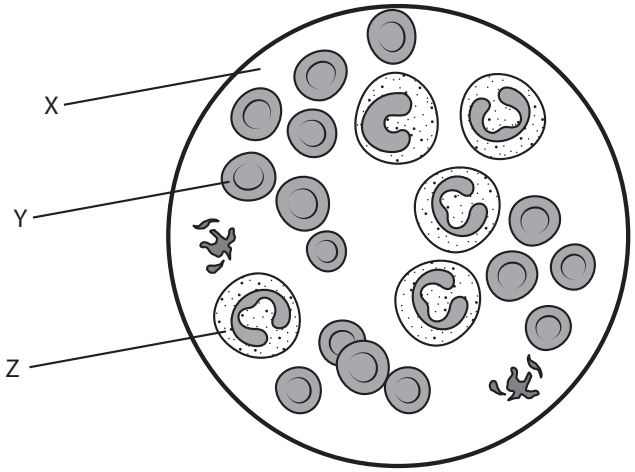
Further practice · Attempt part b only.

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2 (a) The diagram shows components, X, Y and Z, in a sample of human blood.



(i) Name components X, Y and Z.

(3)

X .....

Y .....

Z .....

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(ii) Describe the function of component Y.

(2)

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(iii) A sample of blood from a patient is analysed in a hospital.

A scientist finds that the number of component Z in the blood sample is higher than normal.

State a possible reason for the increase in the number of component Z.

(1)

.....

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4. Question 2 · part b · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b only.

(b) Blood is transported around the body inside blood vessels.

Draw **one** straight line from each blood vessel to a feature of the blood vessel.

(2)

Blood vessel	Feature of blood vessel
artery	contains valves to prevent backflow of blood
vein	has muscular walls to maintain high blood pressure
capillary	has thin walls for exchange of materials

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4. Question 2 · part b · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b only.

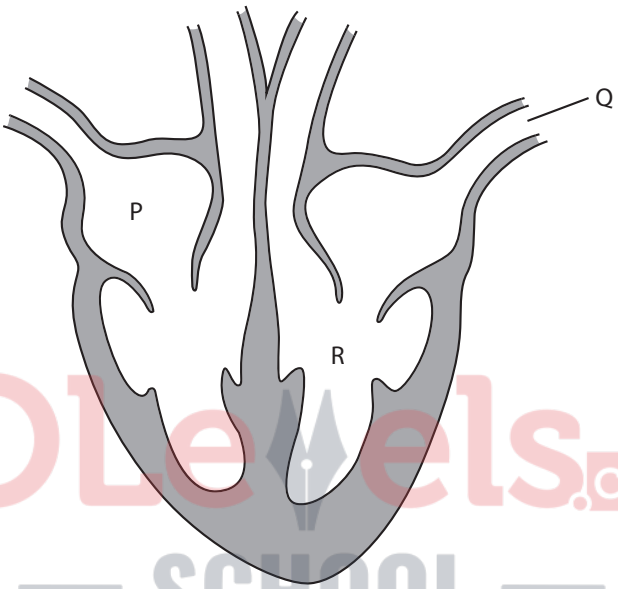
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(c) The heart pumps blood around the body.

The diagram shows a human heart.



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(i) The heart is divided into two halves.

Each half is divided into two chambers.

Name chamber R.

(1)

(ii) Describe the route that the blood travels from part P to part Q.

(4)

(Total for Question 2 = 13 marks)

4. Question 2 · part b · 2 marks · Working and answer

Working and answer

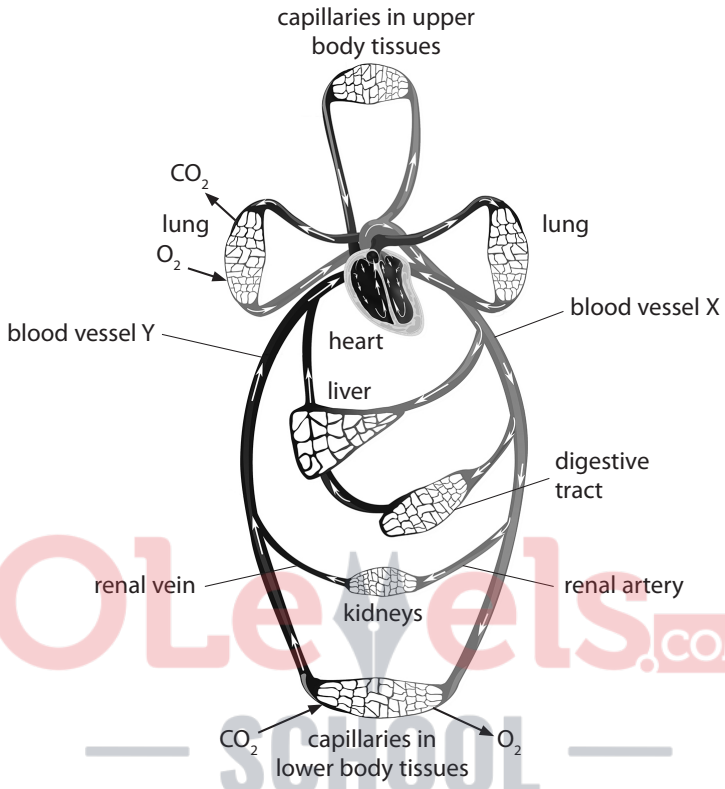
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5. Question 6 · part b · 4 marks

June 2019 Paper 2

Further practice · Attempt part b only.

6 The diagram shows the human circulatory system.



(Source: © Designua/Shutterstock)

(a) (i) What is the name of blood vessel X?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

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5. Question 6 · part b · 4 marks

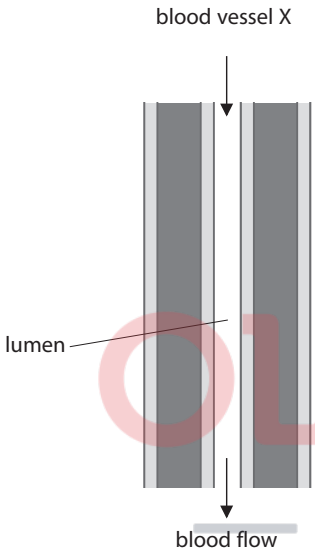
June 2019 Paper 2

Further practice · Attempt part b only.

(iii) The diagram shows a longitudinal section through blood vessel X.

Draw a diagram to show a longitudinal section through blood vessel Y.

(3)



(iv) Explain the differences in the structures of blood vessel X and blood vessel Y.

(3)

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5. Question 6 · part b · 4 marks

June 2019 Paper 2

Further practice · Attempt part b only.

(b) Explain two adaptations of capillaries that increase the rate of diffusion of substances into body tissues.

(4)

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(Total for Question 6 = 12 marks)

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5. Question 6 · part b · 4 marks · Working and answer

Working and answer

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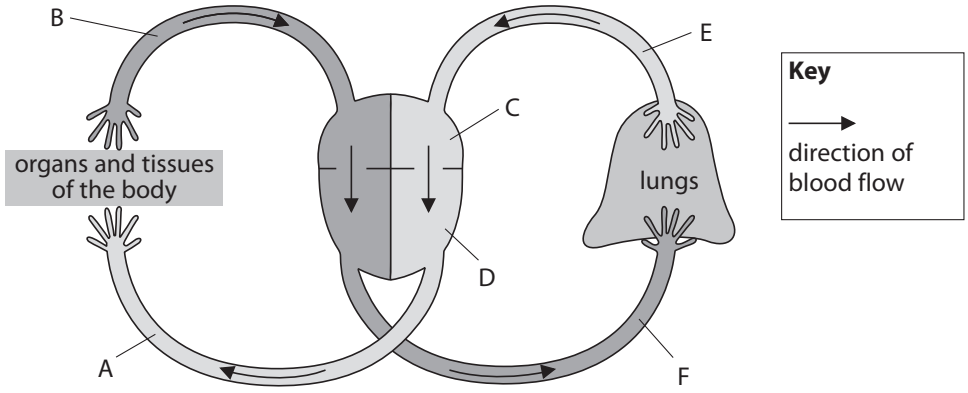
6. Question 4 · part d · 2 marks

January 2020 Paper 1

Further practice · Attempt part d only.

4 William Harvey was the first person to describe the circulation of blood around the body.

The diagram shows a model of the human circulatory system based on the research carried out by Harvey.



(a) Harvey suggested that each side of the heart was divided into two chambers.

State the names of chambers C and D.

(2)

chamber C .....

chamber D .....

(b) Harvey also suggested that blood is transported to and from the heart in different types of vessel.

Explain how the diagram supports Harvey's suggestion.

(4)

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6. Question 4 · part d · 2 marks

January 2020 Paper 1

Further practice · Attempt part d only.

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(c) Before Harvey's discoveries some scientists thought that the blood was circulated around the body by the liver.

State why this idea is not correct.

(1)

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(d) Harvey's research did not include capillaries.

Describe the function of capillaries.

(2)

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(Total for Question 4 = 9 marks)

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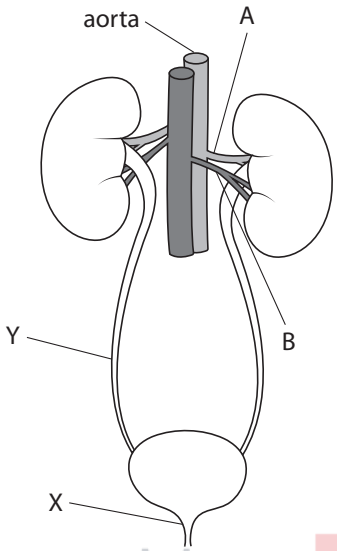
Working and answer

7. Question 8 · part a.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.ii only.

8 The diagram shows part of the human renal system.



(a) (i) Give the names of structures X and Y. **(2)**

structure X .....

structure Y .....

(ii) Describe the function of the aorta. **(2)**

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(iii) Compare the composition of the blood in vessels A and B. **(2)**

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7. Question 8 · part a.ii · 2 marks · Working and answer

Working and answer

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8. Question 5 · part b · 4 marks

June 2025 Paper 2

Further practice · Attempt part b only.

5 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Losartan belongs to a group of drugs known as angiotensin-II receptor antagonists. The body produces angiotensin-II, which is a substance that binds to receptors in blood vessels, causing them to narrow. This causes an increase in blood pressure. Losartan prevents angiotensin-II binding to these receptors, lowering blood pressure.

If high blood pressure is left untreated, a decrease in kidney function can result in proteinuria. This is a condition that causes high levels of protein to be found in a person's urine. People with high blood pressure and a resultant thickening of the left ventricle can be treated with Losartan to reduce the risk of stroke.

People with high blood pressure should not take Losartan if their liver is not functioning properly. Women who are pregnant or who are breast feeding should not take Losartan.

(a) Describe how the action of Losartan differs from the action of an ACE inhibitor. (lines 4 and 5)

(4)

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8. Question 5 · part b · 4 marks

June 2025 Paper 2

Further practice · Attempt part b only.

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(b) Explain how the structure of a named type of blood vessel allows angiotensin-II to increase blood pressure. (lines 2 to 5)

(4)

(c) Explain why proteinuria is a sign of problems with kidney function. (lines 6 to 8)

(2)

(d) Suggest why high blood pressure causes the left ventricle to become thickened. (lines 8 to 9)

(2)

(e) Explain why people who have high blood pressure are advised not to take Losartan if their liver is not functioning properly. (lines 10 to 11)

(2)

(Total for Question 5 = 14 marks)

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8. Question 5 · part b · 4 marks · Working and answer

Working and answer

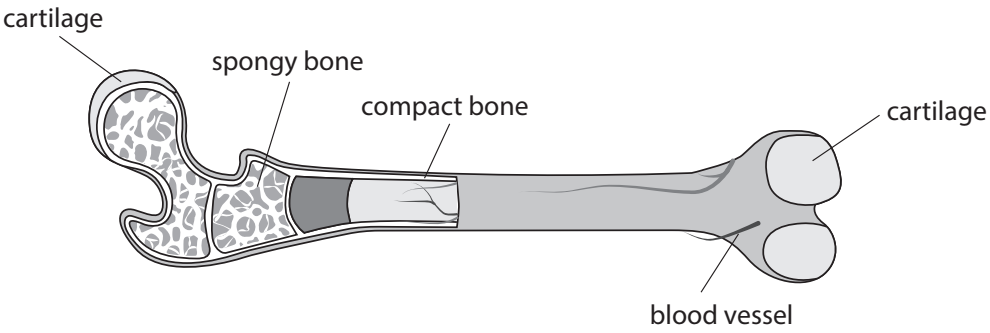
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9. Question 1 · part a.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.iii only.

1 (a) The diagram shows a bone found in the human body.



(Source: adapted from: © Designua / Shutterstock)

(i) State the name of the type of bone shown in the diagram.

(1)

.....

(ii) Place one tick (✓) in each row to give the function of each part of the bone.

(3)

Part	Function		
	reduces friction	where red blood cells are produced	provides strength
compact bone			
spongy bone			
cartilage			

(iii) Describe the function of the blood vessel in the bone.

(2)

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9. Question 1 · part a.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.iii only.

(b) Bones are moved by muscles.

(i) Use the words in the box to complete the passage about bones and muscles.

(4)

cartilage	contracts	ligaments	lowers	pulls
pushes	raises	relaxes	tendons	

Muscles are attached to bones by ..... .

To raise the lower arm, the biceps ..... and the triceps ..... .

The biceps ..... on the radius, which raises the lower arm.

(ii) Energy is transferred by the biceps muscle when raising the lower arm.

The biceps muscle applies a force of 125 N to raise the lower arm a distance of 0.50 m.

Use this formula to calculate the energy transferred by the biceps muscle when raising the lower arm 0.50 m.

energy transferred (J) = force (N) × distance (m)

(2)

energy transferred = ... J

(Total for Question 1 = 12 marks)

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9. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

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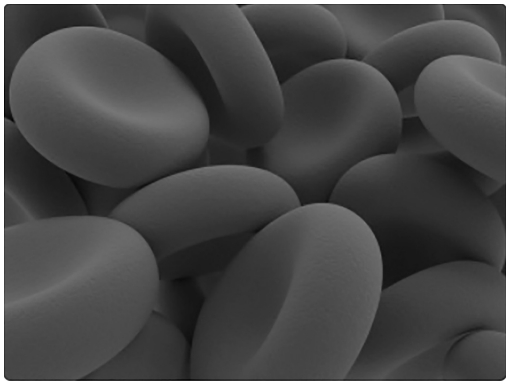
10. Question 1 · part b.i · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.i only.

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1 (a) The photograph shows a group of red blood cells.



(Source: RomanenkoAlexey/Shutterstock)

(i) Describe the function of red blood cells.

(3)

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(ii) Explain how the structure of the red blood cell is adapted to its function.

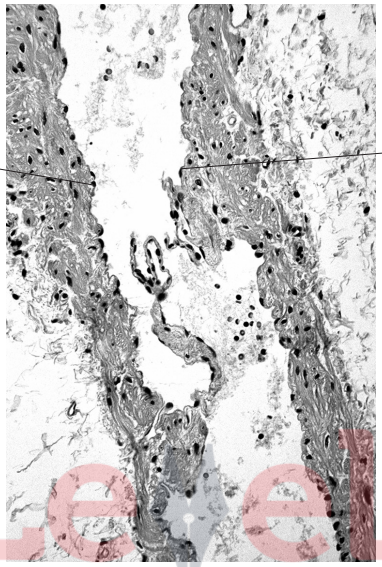
(4)

10. Question 1 · part b.i · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.i only.

(b) The photograph shows a section through a blood vessel in a leg.



magnification ×10

(Source: © DR. GLADDEN WILLIS, VISUALS UNLIMITED/SCIENCE PHOTO LIBRARY)

(i) State the type of blood vessel shown in the photograph.

(1)

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(ii) Explain your choice of blood vessel.

(2)

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10. Question 1 · part b.i · 1 mark · Working and answer

Working and answer

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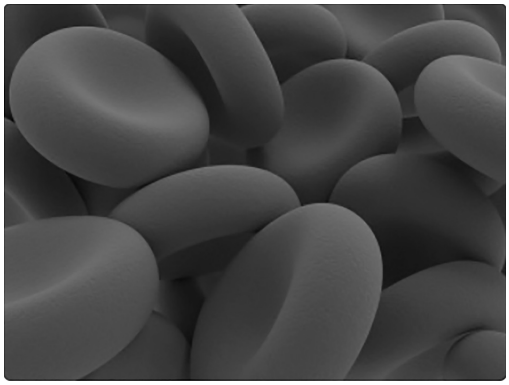
11. Question 1 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

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1 (a) The photograph shows a group of red blood cells.



(Source: RomanenkoAlexey/Shutterstock)

(i) Describe the function of red blood cells.

(3)

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(ii) Explain how the structure of the red blood cell is adapted to its function.

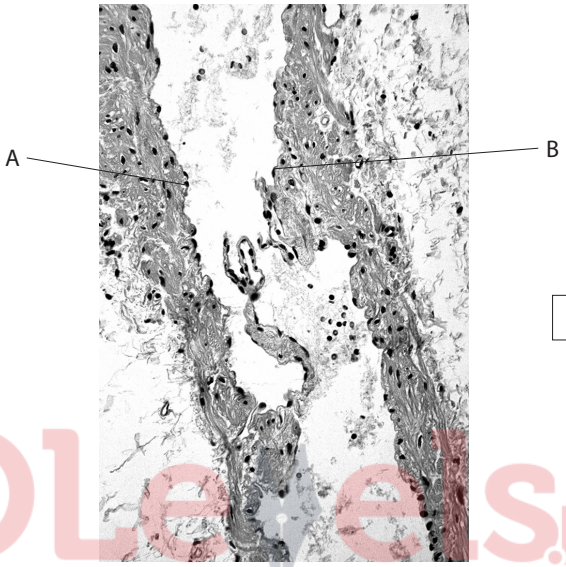
(4)

11. Question 1 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

(b) The photograph shows a section through a blood vessel in a leg.



magnification ×10

(Source: © DR. GLADDEN WILLIS, VISUALS UNLIMITED/SCIENCE PHOTO LIBRARY)

(i) State the type of blood vessel shown in the photograph.

(1)

(ii) Explain your choice of blood vessel.

(2)

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11. Question 1 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

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(iii) Calculate the actual distance between points A and B.

(2)

distance = ... mm

(Total for Question 1 = 12 marks)

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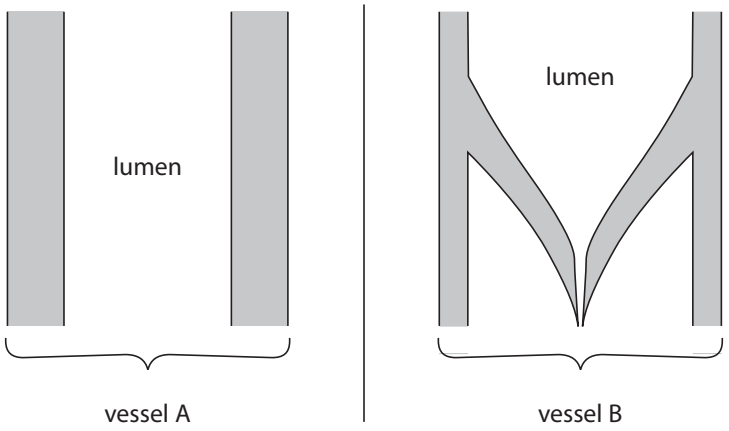
Working and answer

12. Question 3 · part a.ii · 4 marks

June 2022 Paper 2R

Further practice · Attempt part a.ii only.

3 (a) The diagram shows two types of blood vessel, A and B.



(i) Draw an arrow on the diagram to show the direction of blood flow in vessel B.

(1)

(ii) Identify each type of blood vessel, giving explanations for why there are differences in their structures.

(4)

Vessel A

Vessel B

(iii) State the name of one other type of blood vessel.

(1)

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12. Question 3 · part a.ii · 4 marks · Working and answer

Working and answer

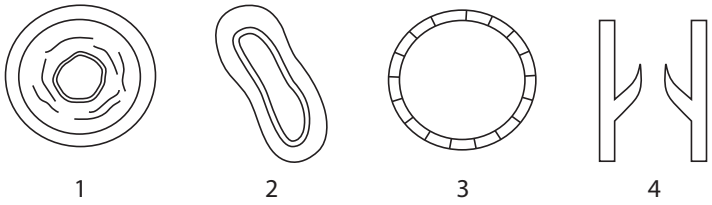
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13. Question 6 · part a.i · 4 marks

November 2024 Paper 2

Further practice · Attempt part a.i only.

6 (a) The diagram shows different sections through various blood vessels.



Not to scale

(i) Identify each type of blood vessel.

(4)

blood vessel 1

blood vessel 2

blood vessel 3

blood vessel 4

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13. Question 6 · part a.i · 4 marks

November 2024 Paper 2

Further practice · Attempt part a.i only.

(ii) State one reason for your choice for each blood vessel.

(4)

blood vessel 1

blood vessel 2

blood vessel 3

blood vessel 4

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13. Question 6 · part a.i · 4 marks · Working and answer

Working and answer

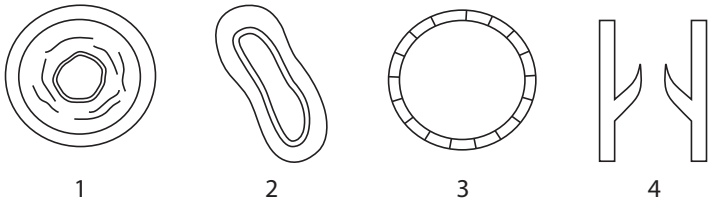
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14. Question 6 · part a.ii · 4 marks

November 2024 Paper 2

Further practice · Attempt part a.ii only.

6 (a) The diagram shows different sections through various blood vessels.



Not to scale

(i) Identify each type of blood vessel.

(4)

blood vessel 1

blood vessel 2

blood vessel 3

blood vessel 4

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14. Question 6 · part a.ii · 4 marks

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Further practice · Attempt part a.ii only.

(ii) State one reason for your choice for each blood vessel.

(4)

blood vessel 1

blood vessel 2

blood vessel 3

blood vessel 4

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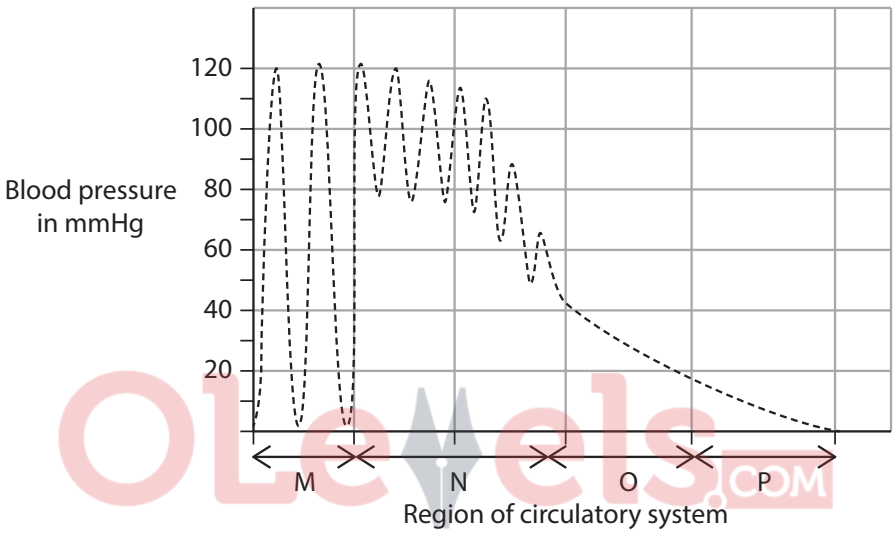
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14. Question 6 · part a.ii · 4 marks

November 2024 Paper 2

Further practice · Attempt part a.ii only.

(b) The graph shows changes in blood pressure in four regions of the circulatory system labelled M, N, O and P.



Identify each region, M, N, O and P, of the circulatory system.

(4)

M .....

N .....

O .....

P .....

(Total for Question 6 = 12 marks)

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14. Question 6 · part a.ii · 4 marks · Working and answer

Working and answer

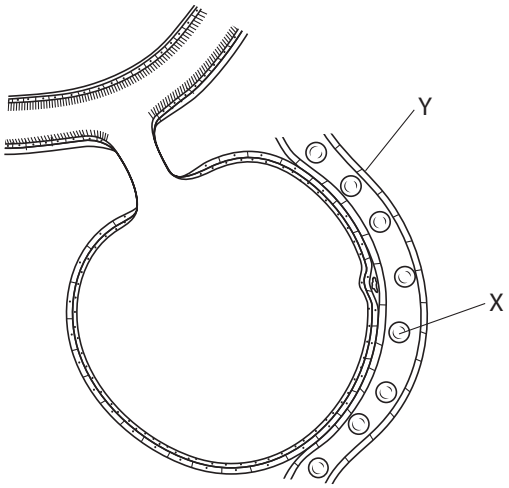
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15. Question 3 · part b · 2 marks

November 2020 Paper 2

Further practice · Attempt part b only.

3 The diagram shows an alveolus and its blood supply.



(a) (i) What is the name of blood component X? **(1)**

- ☒ **A** plasma
- ☒ **B** platelet
- ☒ **C** red blood cell
- ☒ **D** white blood cell

(ii) State three ways in which air in the alveolus differs from air in the atmosphere. **(3)**

1 .....

.....

2 .....

.....

3 .....

.....

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15. Question 3 · part b · 2 marks

November 2020 Paper 2

Further practice · Attempt part b only.

(iii) Give three features of alveoli that allow efficient gas exchange.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(b) Structure Y is a capillary.

Give two features in the diagram which show that structure Y is a capillary.

(2)

1 .....

.....

2 .....

.....

(c) Some people have a condition known as emphysema. One symptom of emphysema is the breakdown of elastic fibres in the lung tissue.

Suggest the effects that loss of elastic fibres have on a person's ability to breathe.

(2)

.....

.....

.....

.....

(Total for Question 3 = 11 marks)

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15. Question 3 · part b · 2 marks · Working and answer

Working and answer

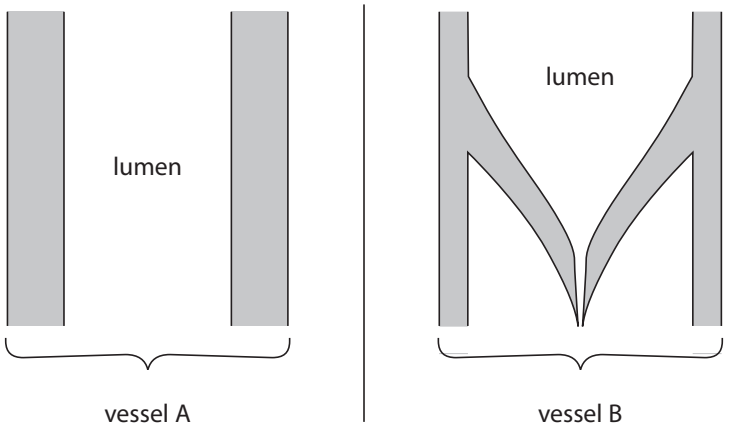
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16. Question 3 · part a.i · 1 mark

June 2022 Paper 2R

Further practice · Attempt part a.i only.

3 (a) The diagram shows two types of blood vessel, A and B.



(i) Draw an arrow on the diagram to show the direction of blood flow in vessel B.

(1)

(ii) Identify each type of blood vessel, giving explanations for why there are differences in their structures.

(4)

Vessel A

Vessel B

(iii) State the name of one other type of blood vessel.

(1)

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16. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

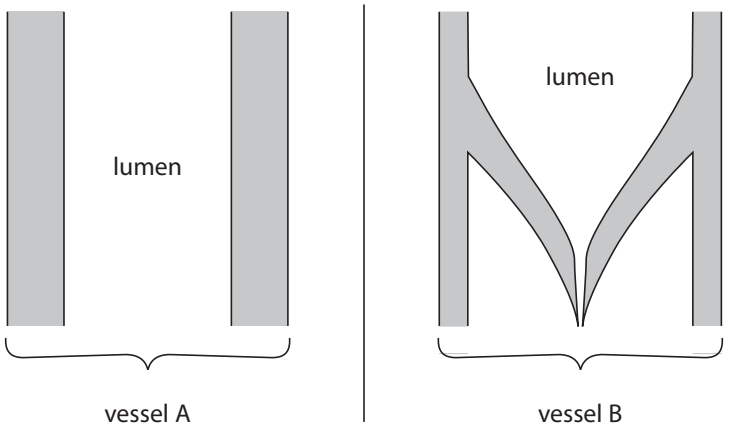
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17. Question 3 · part a.iii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part a.iii only.

3 (a) The diagram shows two types of blood vessel, A and B.



(i) Draw an arrow on the diagram to show the direction of blood flow in vessel B.

(1)

(ii) Identify each type of blood vessel, giving explanations for why there are differences in their structures.

(4)

Vessel A

Vessel B

(iii) State the name of one other type of blood vessel.

(1)

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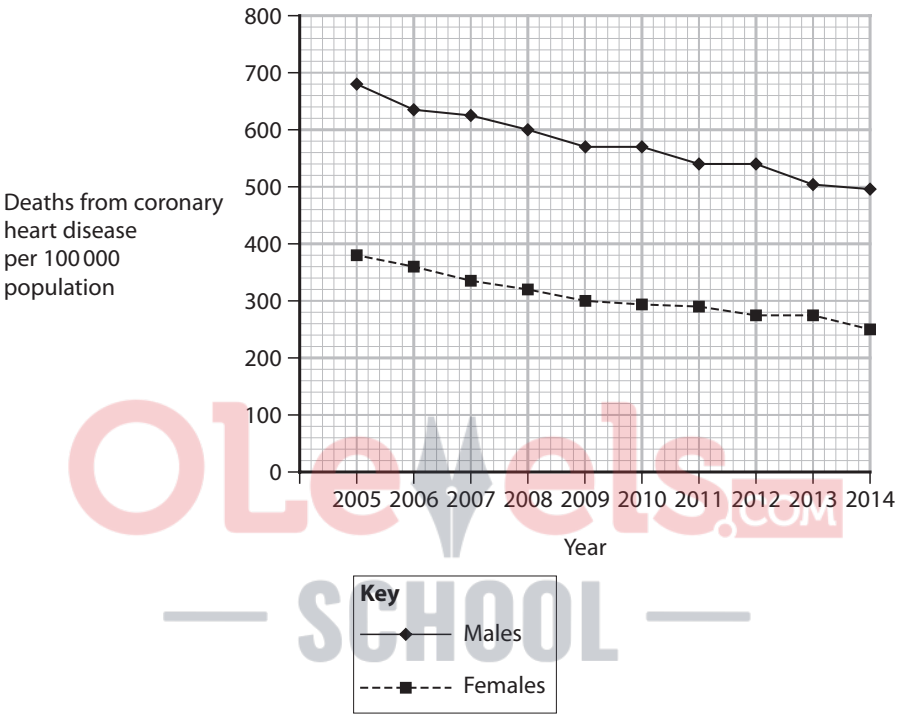
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17. Question 3 · part a.iii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part a.iii only.

(b) Coronary heart disease kills many people each year. The graph shows the number of deaths per 100 000 of the whole population in a part of the world.



(i) Describe the trends shown by the graph.

(3)

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17. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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